

Week 11: Centralized Logging (Part 1)

Date: May 18, 2026

Study Time: 5-8 hours total

GOAL

Begin building your centralized logging infrastructure

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 7: System Logging
 - Focus on: syslog, rsyslog, journald

Video Learning

- ☐ LearnLinuxTV - Search: "Centralized Logging"
 - <https://youtube.com/@LearnLinuxTV>
- ☐ Professor Messer - Search: "Syslog"
 - <https://youtube.com/@professormesser>

Concepts to Understand

- ☐ What is syslog?
 - ☐ rsyslog vs syslogd vs syslog-ng
 - ☐ journald (systemd's logging)
 - ☐ Log severity levels (emerg, alert, crit, err, warning, notice, info, debug)
 - ☐ Log facilities (auth, cron, daemon, kern, mail, user, etc.)
 - ☐ Remote logging (why centralize?)
 - ☐ TCP vs UDP for log forwarding
 - ☐ Log file locations (/var/log/)
-

LAB WORK (3-4 hours)

Understanding Current Logging

☐ View system logs

bash

```
# On systemd systems (Ubuntu 16.04+, Fedora):  
journalctl  
  
# View last 50 lines  
journalctl -n 50  
  
# Follow in real-time  
journalctl -f  
  
# View today's logs  
journalctl --since today  
  
# View specific service  
journalctl -u sshd
```

☐ Traditional log files

bash

```
# Common log locations:  
ls -lh /var/log/  
  
# System messages  
tail -f /var/log/messages # Fedora/RHEL  
tail -f /var/log/syslog   # Ubuntu/Debian  
  
# Authentication logs  
tail -f /var/log/auth.log # Ubuntu  
tail -f /var/log/secure   # Fedora  
  
# Kernel messages  
dmesg  
tail -f /var/log/kern.log
```

Set Up Centralized Logging Server (Fedora)

Your Fedora server will receive logs from all other devices.

☐ On Fedora Server: Install rsyslog (usually pre-installed)

bash

```
ssh fedora

# Check if installed
systemctl status rsyslog

# Install if needed
sudo dnf install rsyslog
```

☐ Configure rsyslog to receive remote logs

bash

```
sudo nano /etc/rsyslog.conf

# Find these lines (they're usually commented out):
# Provides UDP syslog reception
#module(load="imudp")
#input(type="imudp" port="514")

# Provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# UNCOMMENT them (remove #):
module(load="imudp")
input(type="imudp" port="514")

module(load="imtcp")
input(type="imtcp" port="514")

# Save and exit (Ctrl+X, Y, Enter)
```

☐ Create template for remote logs (optional but recommended)

bash

```
sudo nano /etc/rsyslog.conf

# Add before the GLOBAL DIRECTIVES section:

# Template for remote logs
$template RemoteHost, "/var/log/remote/%HOSTNAME%/%PROGRAMNAME%.log"
*. * ?RemoteHost

# This creates separate directories for each host
```

☐ Restart rsyslog

bash

```
sudo systemctl restart rsyslog

# Verify it's running
systemctl status rsyslog
```

☐ Open firewall port 514

bash

```
# Fedora (firewalld):
sudo firewall-cmd --permanent --add-port=514/tcp
sudo firewall-cmd --permanent --add-port=514/udp
sudo firewall-cmd --reload

# Verify
sudo firewall-cmd --list-all
```

☐ Verify rsyslog is listening

bash

```
ss -tulpn | grep 514

# Should show rsyslogd listening on port 514
```

Configure ONE Client to Forward Logs (Pi Device)

Start with just ONE device to test. We'll add the rest next week.

☐ On Raspberry Pi 4: Configure log forwarding

bash

```
ssh pi4

# Backup config first
sudo cp /etc/rsyslog.conf /etc/rsyslog.conf.backup

# Edit config
sudo nano /etc/rsyslog.conf

# Add at the END of the file:
# Forward all logs to Fedora server
*. * @192.168.1.100:514

# Note: @ means UDP, @@ means TCP
# Replace 192.168.1.100 with your Fedora's Tailscale IP

# For TCP (more reliable):
*. * @@100.64.0.10:514

# Save and exit
```

☐ Restart rsyslog on Pi

bash

```
sudo systemctl restart rsyslog
systemctl status rsyslog
```

Test Log Forwarding

☐ Send a test message from Pi

bash

```
# On Pi 4:
logger "TEST MESSAGE FROM PI4 - If you see this, logging works!"
```

☐ Check logs on Fedora server

bash

```
# On Fedora:
ssh fedora

# View messages log
sudo tail -f /var/log/messages
# Should see your test message!

# If using the template:
sudo tail -f /var/log/remote/pi4/logger.log
```

☐ Generate different types of logs

bash

```
# On Pi 4:

# Auth log (SSH attempt)
ssh localhost # Enter wrong password

# System log
logger -p user.info "Info level test message"

# Error log
logger -p user.err "Error level test message"

# Cron log
logger -t cron "Cron test message"
```

☐ Verify all appear on Fedora

bash

```
# On Fedora:
sudo tail -f /var/log/messages
# Watch for your test messages
```

Troubleshooting

If logs aren't appearing on Fedora:

☐ Check rsyslog is running on both machines

```
bash
```

```
systemctl status rsyslog
```

☐ Check firewall allows port 514

```
bash
```

```
# On Fedora:  
sudo firewall-cmd --list-all | grep 514
```

```
# Or test with telnet  
telnet fedora-ip 514
```

☐ Check for errors in rsyslog

```
bash
```

```
journalctl -u rsyslog -n 50
```

☐ Verify network connectivity

```
bash
```

```
# From Pi:  
ping fedora-tailscale-ip
```

☐ Check SELinux (Fedora)

```
bash
```

```
# Temporarily disable to test  
sudo setenforce 0  
  
# If this fixes it, you need to adjust SELinux policy  
# Re-enable after testing:  
sudo setenforce 1
```

Understanding Log Locations

☐ Map out where logs go

```
bash
```

```
# On Fedora, explore:
ls -lh /var/log/

# Common files:
# messages - General system messages
# secure - Authentication/security
# cron - Cron job logs
# maillog - Email system
# audit/audit.log - Security auditing

# If using template, remote logs:
ls -lh /var/log/remote/
ls -lh /var/log/remote/pi4/
```

FLASHCARD COMMANDS

Create flashcards for these commands:

1. `rsyslogd`
2. `/etc/rsyslog.conf`
3. `journalctl`
4. `journalctl -f`
5. `journalctl -u SERVICE`
6. `logger`
7. `logger -p FACILITY.LEVEL`
8. `tail -f /var/log/messages`
9. `systemctl restart rsyslog`
10. `dmesg`

Concepts for Flashcards:

- What is rsyslog?
- What's the difference between @ and @@ in rsyslog?
- What port does syslog use?
- What are the 8 severity levels?
- Where do logs go by default?
- What is journald?
- Difference between rsyslog and journald?

Log Severity Levels (0-7): 0. emerg - System is unusable

1. alert - Action must be taken immediately
 2. crit - Critical conditions
 3. err - Error conditions
 4. warning - Warning conditions
 5. notice - Normal but significant
 6. info - Informational messages
 7. debug - Debug-level messages
-

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 2.1 Given a scenario, analyze system properties and remediate accordingly
 - 4.1 Given a scenario, analyze and troubleshoot storage issues
 - 4.2 Given a scenario, analyze and troubleshoot network resource issues
 - 4.4 Given a scenario, analyze and troubleshoot application and hardware issues
-

NOTES & REFLECTIONS

Fedora server rsyslog configuration:

- Listening on port: _____ (TCP/UDP)
- Log storage location: _____
- Firewall rule added: _____

Pi 4 forwarding configuration:

- Forwarding to: _____ (IP:Port)
- Protocol: _____ (TCP/UDP)
- Test message received? ☐ Yes ☐ No

Log flow diagram:

```
Pi 4 (100.64.0.30)
  ↓ (port 514)
Fedora Server (100.64.0.10)
  ↓
/var/log/messages (or /var/log/remote/pi4/)
```

Issues encountered:

SELinux issues? (Fedora)

Completed: ☐

QUICK REFERENCE

rsyslog.conf Syntax:

```
FACILITY.LEVEL      ACTION

Examples:
*. * @192.168.1.100:514      # Forward all logs via UDP
*. * @@192.168.1.100:514    # Forward all logs via TCP
*.info /var/log/info.log    # Log info and above to file
auth.* /var/log/auth.log    # All auth logs to file
kern.crit /var/log/kernel-crit # Critical kernel messages
```

Common Facilities:

- auth, authpriv - Security/authentication
- cron - Cron daemon
- daemon - Other daemons
- kern - Kernel messages
- mail - Mail system
- user - User-level messages
- local0-local7 - Custom use

logger Command Examples:

```
bash

# Simple message
logger "This is a test"

# Specify priority
logger -p user.err "This is an error"

# Specify tag
logger -t myapp "Application message"

# All together
logger -p local0.info -t backup "Backup completed successfully"
```

Viewing Logs:

```
bash
```

```
# Live follow
```

```
journalctl -f
```

```
tail -f /var/log/messages
```

```
# Last N lines
```

```
journalctl -n 100
```

```
tail -n 100 /var/log/messages
```

```
# Since specific time
```

```
journalctl --since "2026-05-18 09:00:00"
```

```
journalctl --since "1 hour ago"
```

```
journalctl --since today
```

```
# For specific service
```

```
journalctl -u sshd
```

```
journalctl -u rsyslog
```

```
# By priority
```

```
journalctl -p err
```

```
journalctl -p warning
```

Next Week: You'll add ALL remaining devices to centralized logging!

Previous Week: Firewall Rules

Next Week: Centralized Logging (Part 2)